

Secure Note-Taking API (Flow of application through Screen Shots)

We will create the whole application by following steps:

To start the application install the zip and open the folder in the Visual Studio and press F5. Then the swagger interface to check the api on the website will get open in the browser.

1. User Registration:

Code:- This code snippet is there in the Controllers/[AuthController.cs](#) file

```
[HttpPost("register")]
public async Task<IActionResult> Register(AuthRequest req) {
    if (await _context.Users.AnyAsync(u => u.Username == req.Username))
        return BadRequest(new { message = "Username already exists" });

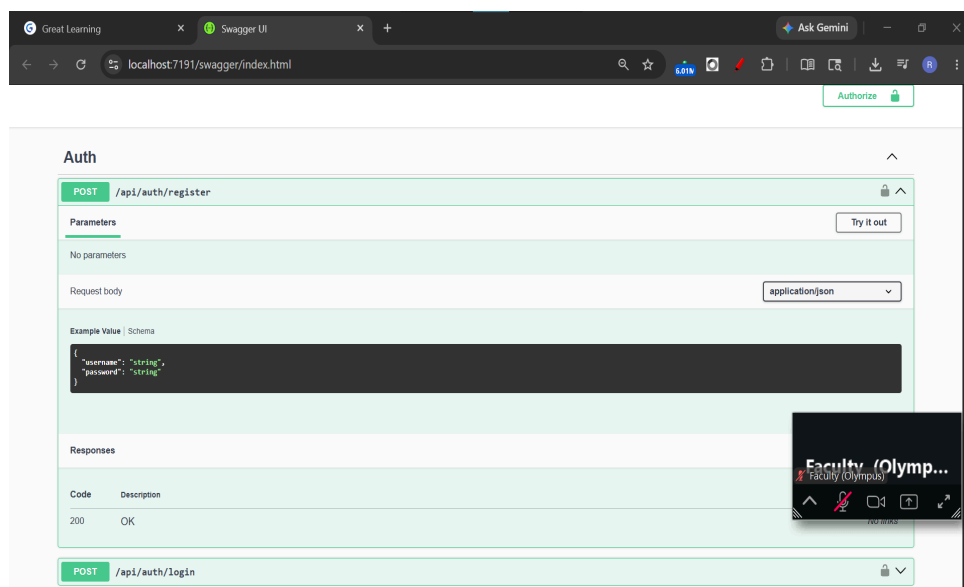
    if (req.Username.Length < 3 || req.Password.Length < 8)
        return BadRequest(new { message = "Username min 3 chars, Password min 8 chars" });

    var user = new User
    {
        Username = req.Username,
        PasswordHash = BCrypt.Net.BCrypt.HashPassword(req.Password)
    };

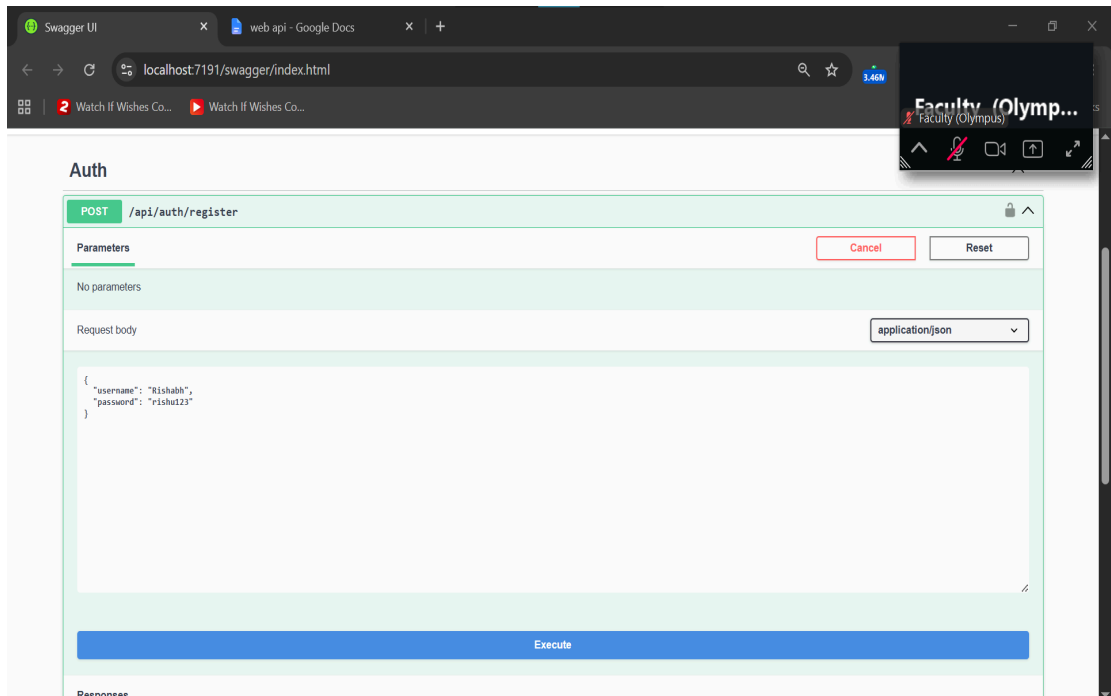
    _context.Users.Add(user);
    await _context.SaveChangesAsync();
    return Ok(new { message = "User registered successfully. Please log in." });
}
```

Screen Shot of the implementation

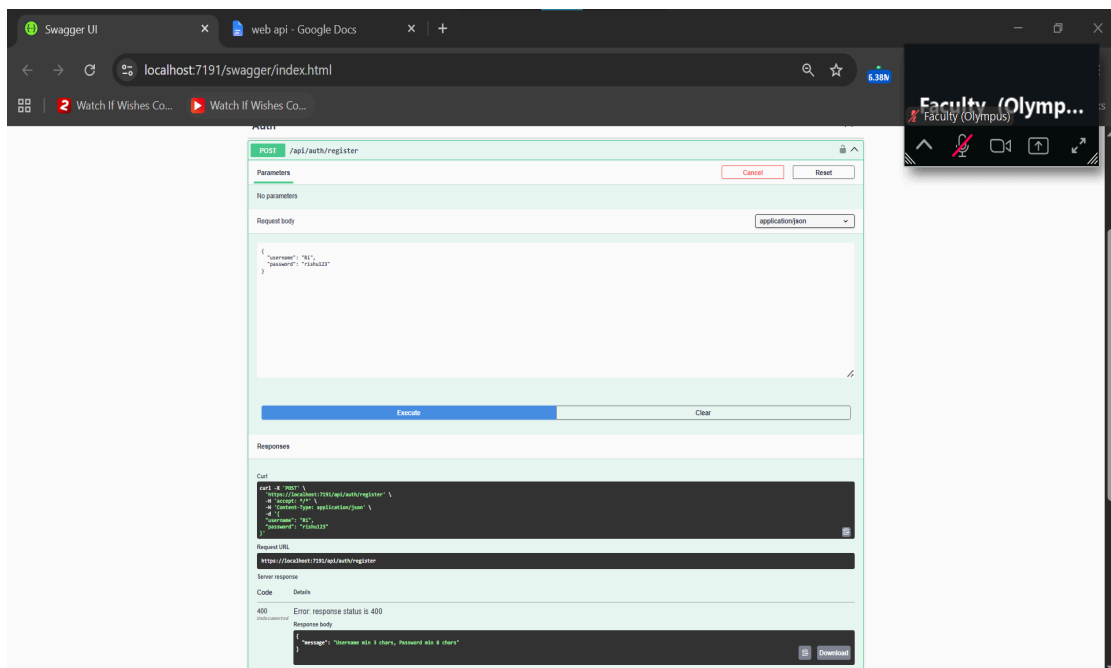
i. In the below ss click the try it out button



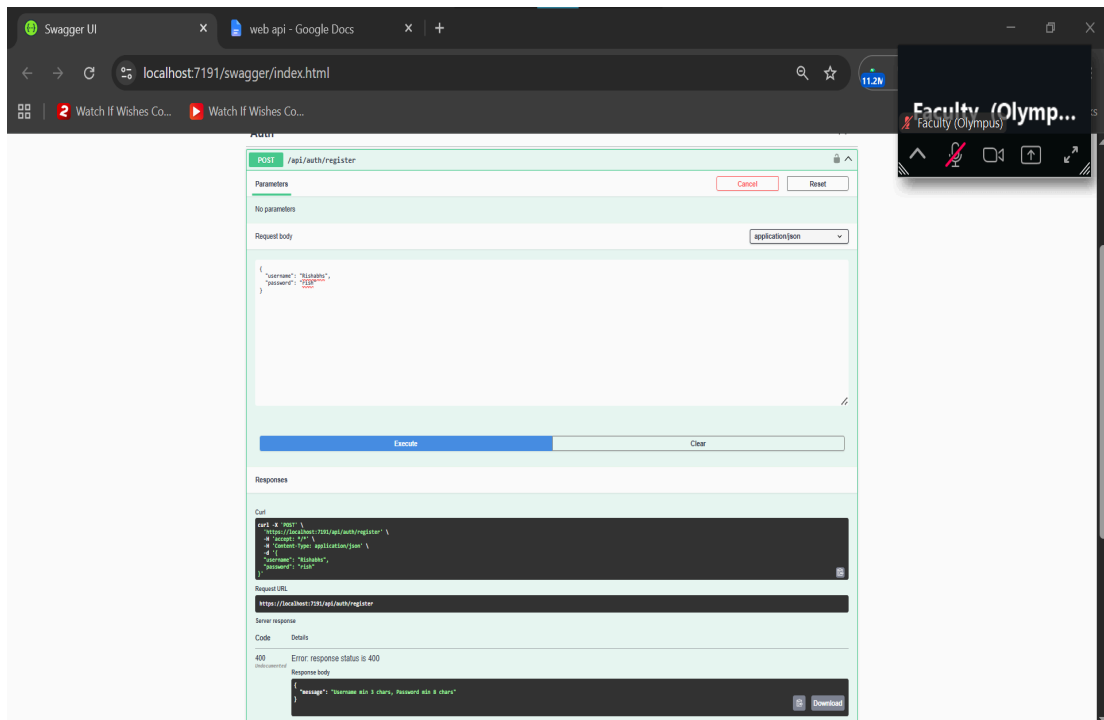
ii. In the below ss we have filled the values in the response body for username and password and now for registering click the “Execute” blue button.



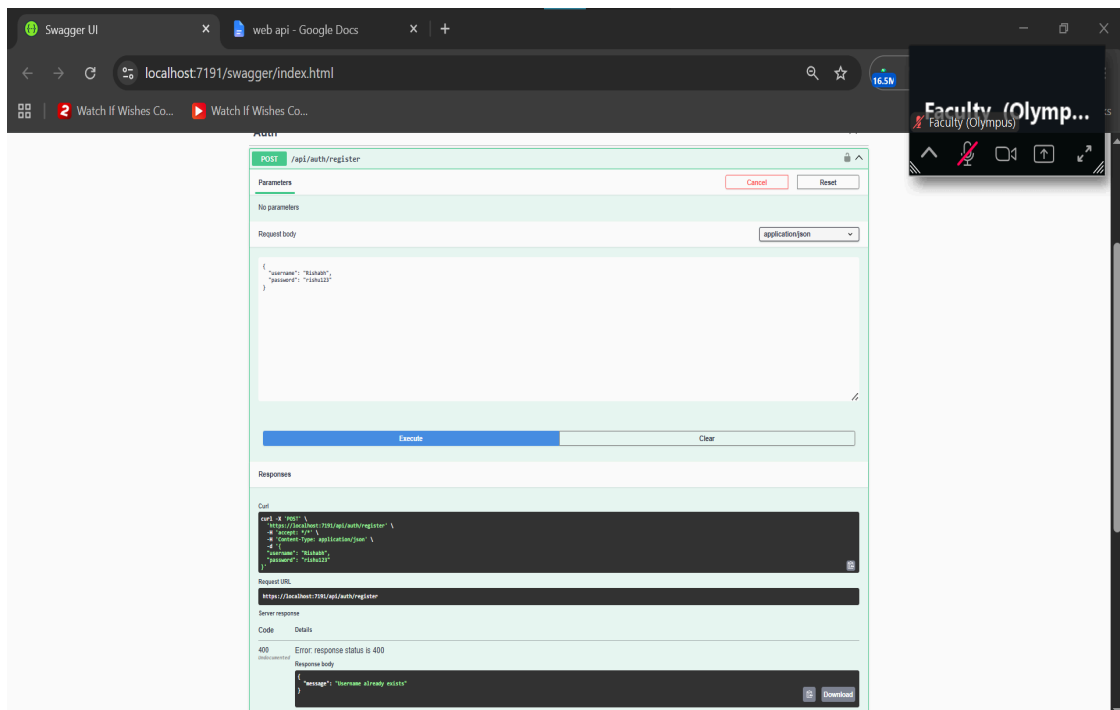
iii. Make sure that the username length must be at least of size 3.



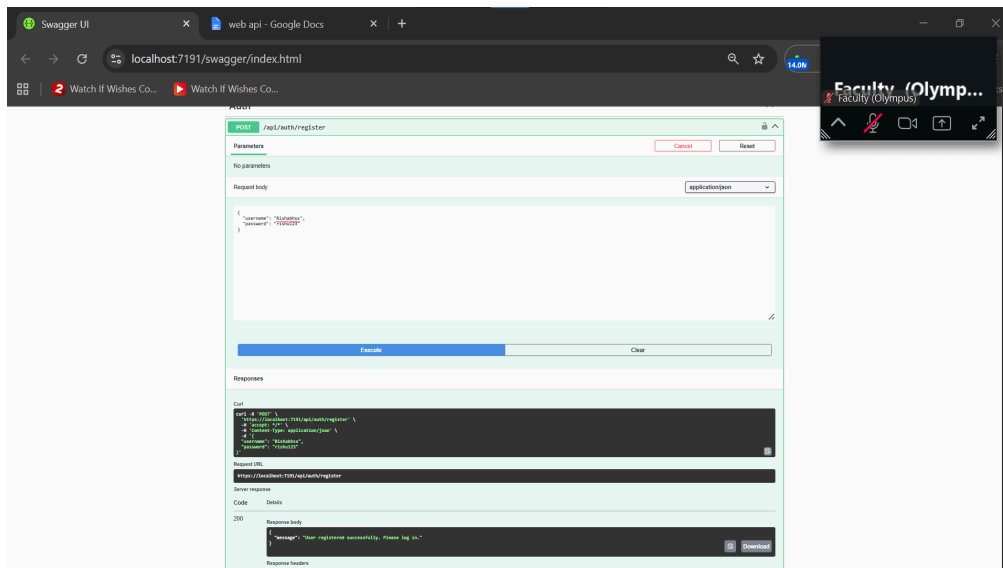
iv. Password must be of at least size 8.



v. Username must be unique



vi. The user got registered successfully.

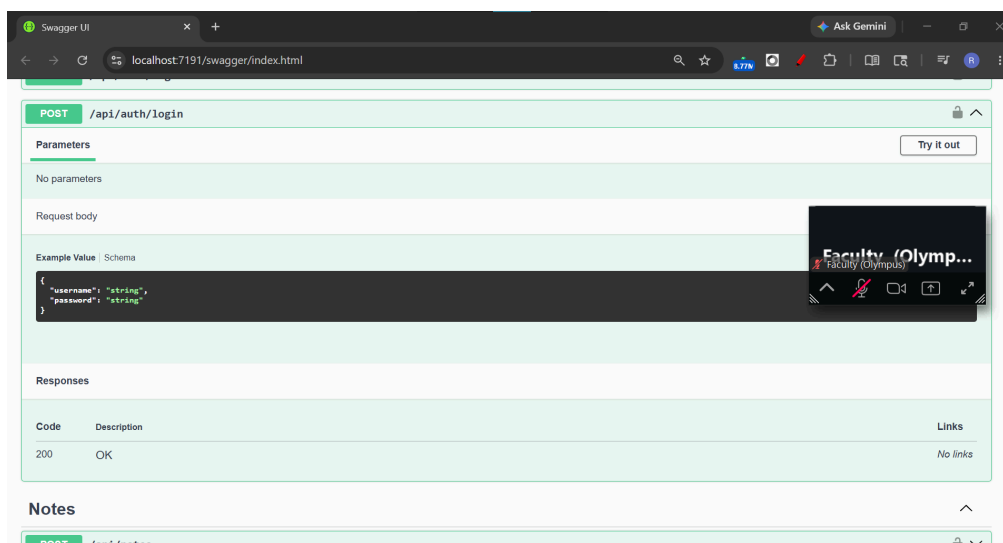


2. User login

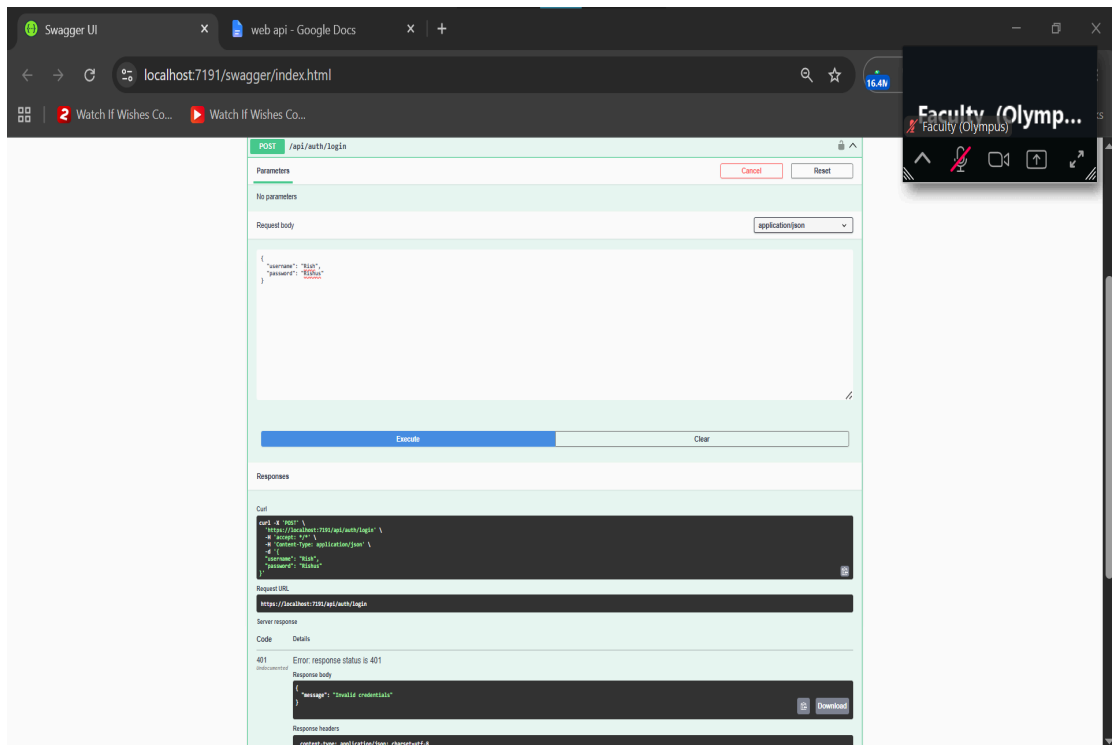
```
[HttpPost("login")]
public async Task<ActionResult> Login(AuthRequest req) {
    var user = await _context.Users.FirstOrDefaultAsync(u => u.Username == req.Username);
    if (user == null || !BCrypt.Net.BCrypt.Verify(req.Password, user.PasswordHash))
        return Unauthorized(new { message = "Invalid credentials" });

    var token = _jwt.GenerateToken(user.Username, user.Id);
    return Ok(new { token, expires_in = 3600, user = new { username = user.Username } });
}
```

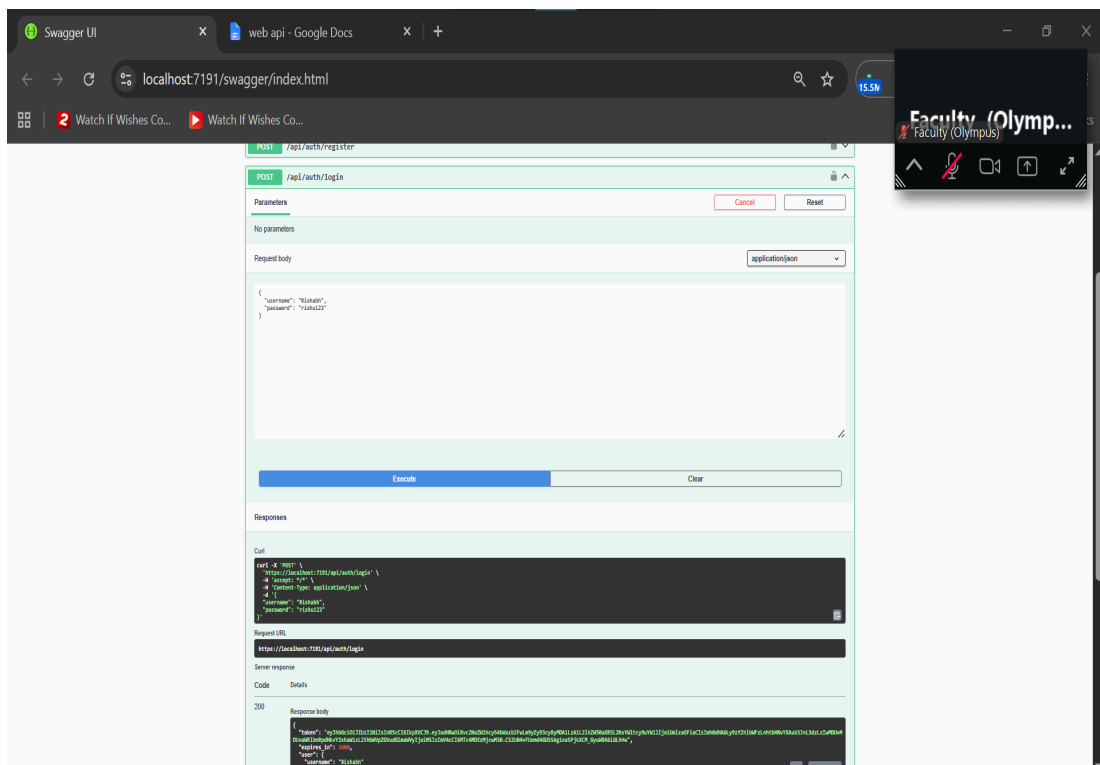
i. In the below ss click the try it out button



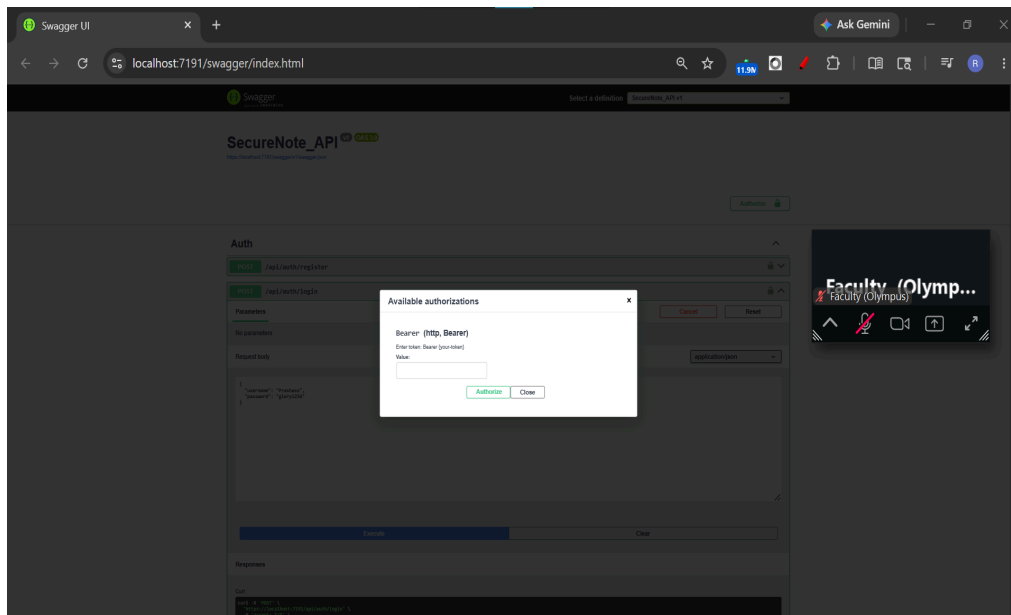
ii. One must enter the valid credentials while login or else it will give error.



iii. In the below ss fill the username and password same as that were used while registering. And copy the token that came inside the Response body as it will be used to create the notes while authorizing the user



iv. Authoring the User Click the Authorize button at the top of the page. And paste the token that you have copied from the response body of login.

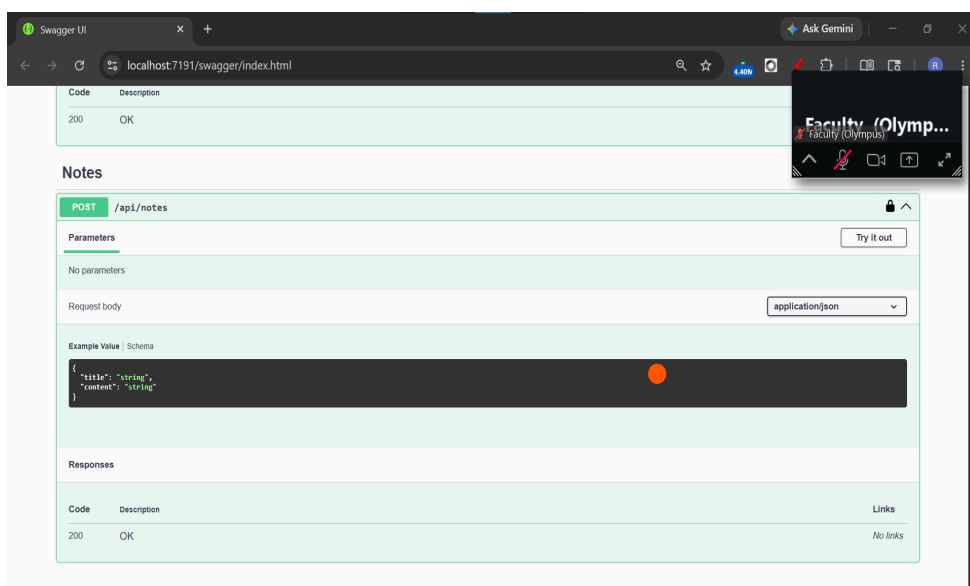


3. Create Notes

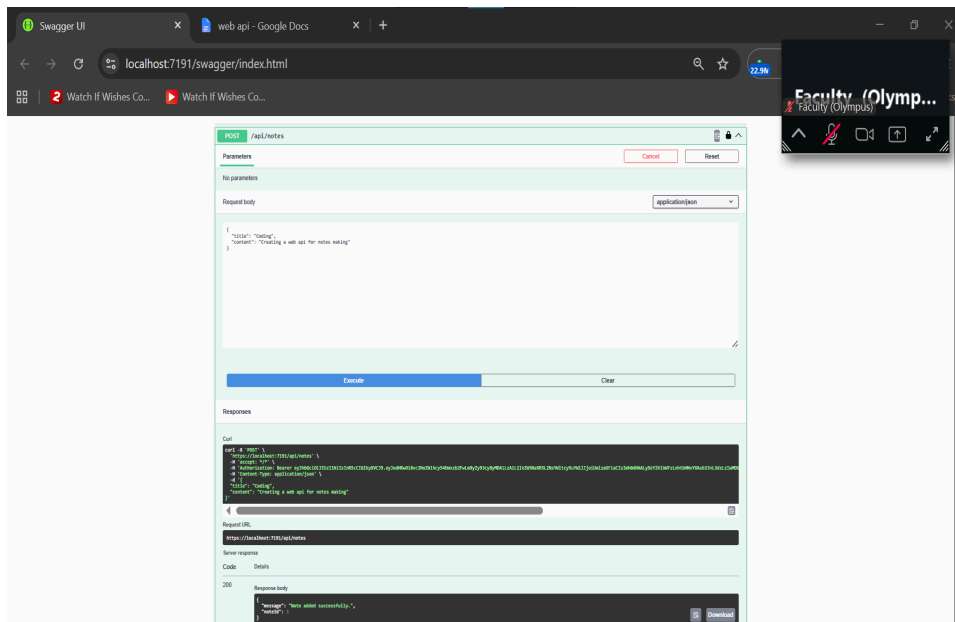
[HttpPost]

```
public async Task<ActionResult> AddNote(NoteRequest req) {  
    var note = new Note { Title = req.Title, Content = req.Content, UserId = GetUserId() };  
    _context.Notes.Add(note);  
    await _context.SaveChangesAsync();  
    return Ok(new { message = "Note added successfully.", noteId = note.Id });  
}
```

i. For creating a note click Try it out in below image



ii. Below ss shows the note successfully got created



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